



### Amazon Pharmacy

Amazon has launched a new service: Amazon Pharmacy, offering home delivery for drugs. <https://digit.in/dec20-33>



### Cira-03

A humanoid coronavirus detection robot, built by an Egyptian inventor is currently undergoing trial. <https://digit.in/dec20-34>

conditions of the battery, and are instead based on generally how batteries deteriorate over time, instead of that particular battery. Researchers from the University of Stanford have developed a model that predicts the actual inner conditions of a rechargeable lithium-ion battery, in real time, using sensors to probe electrochemical parameters that are novel to the purposes of estimating battery life. The lithium ion battery for a vehicle can consist of thousands of cells. The lithium ions in each of these cells shuttle back and forth between the electrodes over the charge and discharge cycles. The conventional approaches for predicting battery capacity assumes that the lithium in the electrodes remains the same over time. In reality, portions of the lithium are lost over time, degrading the battery. The new system uses independent sensors and algorithms for each electrode to estimate the amount of lithium left, a totally new approach for predicting battery life.

To account for the unknown period of time over which the capacity for holding a charge reduces, battery manufacturers, particularly for electric vehicles, add in extra capacity as a buffer. This adds costs and materials. Being able to accurately predict the charge holding capacity, and operating the batteries with precision, can lead to smaller batteries as the buffer can be reduced. The model was able to predict the capacities of the batteries within 2 percent of the actual performance during tests. The tech-

nology can be implemented using the existing sensors in electric vehicles, and can potentially be implemented on vehicles already on the road. There is another benefit of being able to accurately model battery performance. Right now, all second hand batteries are in a wide range of conditions. The tech from Stanford can be used to test, certify and standardise second hand batteries, which makes it cheaper and easier to reuse the batteries for other applications – such as extra storage for energy distribution grids.

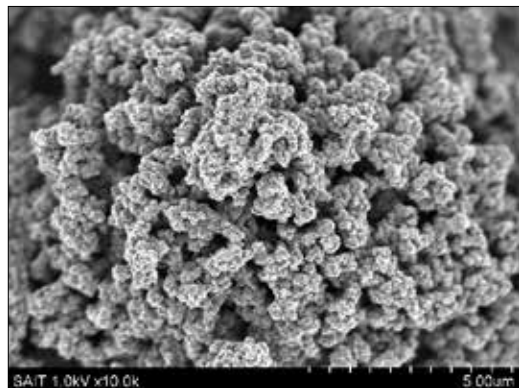
### REUSE IN POWER GRID

What to do with the big influx of second hand batteries used in electric vehicles? This is a big problem that needs to be solved. In electric vehicles, after a battery deteriorates to 80 percent of its original capacity it is no longer considered to be suitable for use in the vehicle. Laptops and smartphones can be used almost till the point where the battery can store no energy at all.

The batteries may be too degraded to be used for the demanding requirements to power a vehicle, but the used batteries may still be good enough to serve as power storage in less demanding applications.

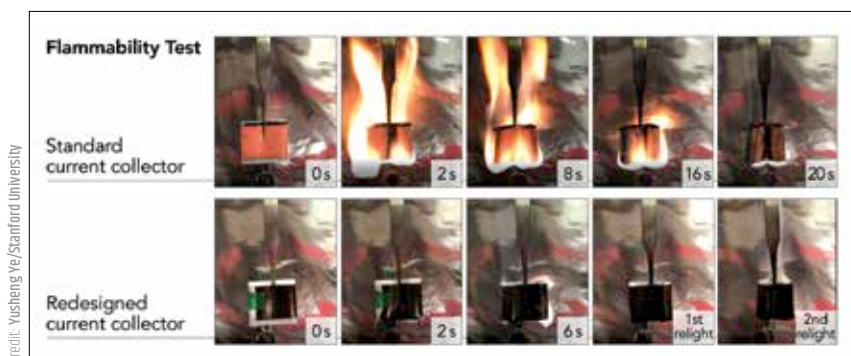
According to researchers from MIT, such batteries could provide backup storage for solar cell farms, for a period of up to ten years. The

study simulated a power plant running on new batteries, and another power plant running on batteries that had deteriorated to 80 percent of their original capacity. They found that the plant using new batteries would not be financially viable,



Pop-corn like "graphene ball" on a charging material that is five times as fast as conventional lithium-ion batteries

but the plant using the deteriorated batteries was, provided the used batteries cost less than 60 percent of the price of new batteries. One of the major roadblocks for implementing this technology is the ability to accurately predict the condition of used batteries, which is where monitoring technologies such as the one mentioned above can play an important role. The study also investigated different deterioration pathways, and found out that charging the battery to only sixty five percent of its capacity, and not allowing it to discharge beyond 15 percent of the capacity achieves the maximum lifetime and value for the battery. This would suggest that those setting up power plants have to resist the tendency to use up as much of the capacity as possible up front, and recover most of their investment in the least time. The study also investigated for how long the batteries could be safely used, and found that in power plants they could be used till the capacity reduced to 60 percent and even beyond. Just figuring out precisely how a battery deteriorates is an active field of research that promises to significantly improve how we use these batteries. 



A flame test with conventional batteries in the top row, and batteries using the new polyimide flame retardant material below